

Online Retail Sales Performance Dashboard

1. Project Overview

The objective of this project was to build an end-to-end Excel sales dashboard using an online retail transaction dataset. The project focused on cleaning raw sales data, preparing it for analysis, creating Pivot Tables, building interactive dashboard visuals, and summarizing key business insights.

The final dashboard provides a high-level view of revenue performance, order volume, customer activity, product performance, country-level sales, and return/cancellation impact.

2. Business Objective

The business needed a clear way to understand sales performance from raw transaction data.

The dashboard was designed to answer these key questions:

Business Question	Dashboard Output
How much revenue was generated?	Total Revenue KPI
How many unique orders were placed?	Unique Orders KPI
How many customers purchased?	Total Customers KPI
What is the average order value?	Average Order Value KPI
Which month had the highest revenue?	Monthly Revenue Trend
Which countries generated the most revenue?	Top Countries Revenue Chart
Which products generated the most revenue?	Top Products Revenue Chart
Who are the highest-value customers?	Top Customers Table
What is the impact of returns/cancellations?	Returns & Cancellations KPI and chart

3. Dataset Description

The dataset contains online retail transaction records.

Each row represents one product line within an invoice. Since one invoice may include multiple products, the same invoice number can appear multiple times.

Column	Description
Invoice	Unique order/invoice identifier
StockCode	Product code
Description	Product name
Quantity	Number of units purchased
InvoiceDate	Date and time of transaction
Price	Unit price
Customer ID	Customer identifier
Country	Customer country

4. Data Cleaning Process

The raw dataset was preserved separately, and all cleaning was performed on a working copy.

Cleaning steps included:

Cleaning Step	Action Taken
Preserved raw data	Kept original dataset unchanged
Converted data into Excel Table	Made formulas and Pivot Tables easier to manage
Formatted fields	Invoice and Customer ID as text, Quantity as number, Price as currency, InvoiceDate as date/time
Created Revenue column	Revenue calculated as $\text{Quantity} \times \text{Price}$

Created Order Status column	Invoices starting with C were flagged as cancelled
Created Transaction Type column	Negative quantities were flagged as returns
Created Analysis Group column	Separated main sales from returns/cancellations
Created date helper fields	Month, Month-Year, Day of Week, and other time fields were prepared
Checked duplicates	Repeated invoice numbers were not treated as duplicates because one invoice can contain multiple product lines
Prepared analysis dataset	Main dashboard focused on valid completed sales

A key cleaning decision was to separate returns and cancellations instead of deleting them. This allowed the dashboard to show clean revenue performance while still tracking return/cancellation impact separately.

5. Data Analysis Process

Pivot Tables were used to summarize the cleaned dataset.

Main analysis areas included:

Analysis Area	Method
Total Revenue	Sum of Revenue
Unique Orders	Distinct count of Invoice
Total Customers	Distinct count / customer-level count
Average Order Value	Total Revenue ÷ Unique Orders
Monthly Revenue	Revenue grouped by month
Country Revenue	Revenue grouped by country
Product Revenue	Revenue grouped by product description

Customer Revenue	Revenue grouped by Customer ID
Returns/Cancellation	Separate analysis using transaction type and analysis group

The dashboard uses Main Sales for core performance metrics and a separate section for Returns & Cancellations.

6. Dashboard Features

The final Excel dashboard includes:

Dashboard Feature	Purpose
KPI Cards	Show executive-level metrics
Monthly Revenue Trend	Shows sales performance over time
Top Countries by Revenue excluding UK	Allows better comparison between non-UK countries
UK Revenue KPI	Separately highlights the dominant revenue country
Top 10 Products by Revenue	Identifies best-performing products
Top 10 Customers by Revenue	Shows highest-value customers
Return/Cancellation Impact Chart	Shows sale vs return activity
Slicers	Allows filtering by country, month, and day of week
Insights Box	Summarizes major findings in plain business language

7. Key Dashboard Metrics

KPI	Value
Total Revenue	\$1,324,642.52
Unique Orders	23,402
Total Customers	4,346
Average Order Value	\$67.70
UK Revenue	\$492,075.42
Return Revenue Impact	\$737,347.86
Returned Quantity	413,574
Return Orders %	19.97%

8. Key Business Insights

Based on the dashboard:

- The United Kingdom generated the majority of revenue.**
UK sales were so dominant that a separate KPI was created for UK revenue, while the country chart excluded the UK to make other countries easier to compare.
- November had the highest monthly revenue.**
Revenue increased strongly toward the end of the year, suggesting possible seasonal demand or holiday shopping behavior.
- A small group of products drove a large share of sales.**
The top product chart shows that a limited number of products contributed significantly to overall revenue.
- Top customers contributed significantly to total revenue.**
The customer table shows that a small number of customers generated high sales value, making them important for retention or loyalty strategies.
- Returns and cancellations should be monitored closely.**
The return/cancellation section shows a notable impact on sales activity, which may affect net revenue and operational efficiency.

9. Business Recommendations

Recommendation	Reason
Focus marketing and inventory planning around peak months	November showed the strongest revenue performance
Monitor UK dependence	The UK dominates revenue, creating market concentration risk
Promote top-performing products	A small set of products drives major revenue
Build loyalty strategies for top customers	High-value customers contribute significantly to sales
Investigate returns and cancellations	Return activity may reduce net revenue and indicate product, delivery, or customer experience issues
Analyze non-UK markets separately	Excluding the UK helps identify growth opportunities in other countries

10. Project Deliverables

The final project includes:

Deliverable	Status
Cleaned dataset	Completed
Data cleaning documentation	Completed
Pivot Tables	Completed
KPI calculations	Completed
Interactive Excel dashboard	Completed
Slicers and filters	Completed
Business insights	Completed
Final report	Completed